🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.1

Determine the Area of Triangles

Lesson 5-2 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

SKY HARBOR MISSION

The Triangular Skylight

You are the lead architect at Sky Harbor Studio. The city has hired your firm to design a glass skylight made of triangular panels for the new transit hub. Before the glass can be cut, you must master the area of triangles — every wrong measurement wastes expensive glass.

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the area of a triangle using the formula A = ½ × base × height.

Language Objective: I can explain my steps using the words base, height, area, and perpendicular.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Area of a Triangle | Base | Height | Area | Perpendicular | Composite figure | Formula | Parallel

👆 *Tap any word to see what it means and a picture.*

1. The space inside a triangle, found with A = ½ × b × h, is the ________.

2. The side of a triangle you measure the height from at a 90° angle is the ________.

3. The perpendicular distance from the base to the opposite vertex is the ________.

4. The amount of flat space inside a triangle is its ________.

5. Two lines that meet at a 90° angle are ________.

6. A shape made of two or more basic shapes combined is a

   ______________________.

7. A math rule written with symbols, like A = ½ × b × h, is a

   ______________________.

8. Two sides that stay the same distance apart and never meet are

   ______________________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How many cans of paint does the homeowner need?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Area of a Triangle** — The space inside a triangle. A triangle is half of a rectangle with the same base and height, so A = ½ × b × h.
  Área de un triángulo — El espacio dentro de un triángulo. Un triángulo es la mitad de un rectángulo con la misma base y altura, por eso A = ½ × b × h.

- [ ] **Base** — Any side of a triangle can be the base. Once you pick a base, the height must be measured perpendicular (at a 90° angle) to it.
  Base — Cualquier lado de un triángulo puede ser la base. Una vez que eliges la base, la altura debe medirse perpendicular (en un ángulo de 90°) a ella.

- [ ] **Height** — The straight-up distance from the base to the top corner.
  Altura — La distancia recta desde la base hasta la esquina de arriba.

- [ ] **Area** — How much space is inside a flat shape.
  Área — Cuánto espacio hay dentro de una figura plana.

- [ ] **Perpendicular** — Two lines that meet to make a square corner (90 degrees).
  Perpendicular — Dos líneas que se unen formando una esquina recta (90 grados).

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The wall's area is ____ sq ft because A = 1/2 × ____ × ____. She needs ____ cans because ____ ÷ 50 = ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The height is the perpendicular distance from the base to the top corner.

Evidence: Students identify the height as the perpendicular (straight-up, 90-degree) distance from the base to the opposite vertex, not the slanted side.

Reasoning: This evidence connects the definition of area to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The wall's area is ____ sq ft because A = 1/2 × ____ × ____. She needs ____ cans because ____ ÷ 50 = ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Area of a Triangle
2. **Notes 2:** Base
3. **Notes 3:** Height
4. **Notes 4:** Area
5. **Notes 5:** Perpendicular
6. **Notes 6:** Composite figure
7. **Notes 7:** Formula
8. **Notes 8:** Parallel
9. **Watch for this mistake:** *A common mistake in Area of Triangles is forgetting the ½ — students multiply base × height like a rectangle and stop there, or they use the triangle's slanted side instead of the perpendicular height that's already given. Always check: did I take half, and did I use the straight-up height, not the slanted side?*
10. **Try It 1:** A. 14 sq ft — *A = ½ × base × height = ½ × 7 × 4 = ½ × 28 = 14 sq ft. Multiply base × height first (28), then take half (14).*
11. **Try It 2:** A. 33 sq ft — *A = ½ × base × height = ½ × 11 × 6 = ½ × 66 = 33 sq ft.*